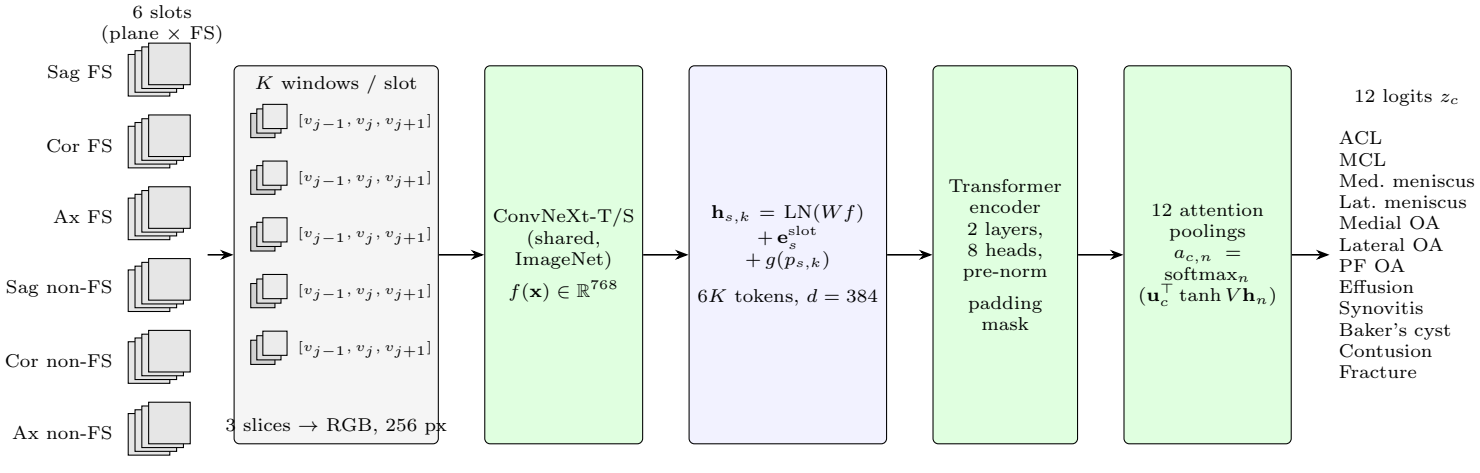

6 slots
(plane × FS)
Sag FS
Cor FS
Ax FS
Sag non-FS
Cor non-FS
Ax non-FS
missing slot ⇒ masked
K windows / slot
$[v_{j-1}, v_j, v_{j+1}]$
$[v_{j-1}, v_j, v_{j+1}]$
$[v_{j-1}, v_j, v_{j+1}]$
$[v_{j-1}, v_j, v_{j+1}]$
$[v_{j-1}, v_j, v_{j+1}]$
3 slices → RGB, 256 px
ConvNeXt-T/S
(shared,
ImageNet)
$f(\mathbf{x}) \in \mathbb{R}^{768}$
$\mathbf{h}_{s,k} = \mathrm{LN}(Wf)$
$+ \mathbf{e}_s^{\mathrm{slot}}$
$+ g(p_{s,k})$
$6K$ tokens, $d = 384$
Transformer
encoder
2 layers,
8 heads,
pre-norm
padding
mask
12 attention
poolings
$a_{c,n} =$
$\mathrm{softmax}_n$
$(\mathbf{u}_c^\top \tanh V\mathbf{h}_n)$
12 logits z_c
ACL
MCL
Med. meniscus
Lat. meniscus
Medial OA
Lateral OA
PF OA
Effusion
Synovitis
Baker's cyst
Contusion
Fracture